NMC-Opt Studio: Scientific Novelty & Architectural Innovations

Academic Research Contributions in Data-Movement-Aware Compiler Optimization

Publication Category: Academic Research Contributions & Innovation Audit
Core Scope: Compiler Design, Intermediate Representation, and Near-Memory Cost Modeling
Repository: https://github.com/MSN-1307/NMC-OPT-CD | **Target:** Heterogeneous CPU/PIM Systems

1. Executive Statement of Novelty

While Processing-In-Memory (PIM) and Near-Memory Computing (NMC) hardware prototypes (such as UPMEM DPU, Samsung HBM-PIM, and SK Hynix AiM) have demonstrated extraordinary bandwidth gains in hardware laboratories, software toolchains remain in their infancy. Existing compilers (e.g., GCC, Clang/LLVM) treat memory accesses as uniform-cost load/store operations, lacking the analytical machinery required to make autonomous offloading decisions between host CPUs and distributed in-memory compute units.

NMC-Opt Studio bridges this fundamental software-hardware gap by introducing an adaptive, data-movement-aware compiler optimization workbench. The platform is not an ad-hoc hardware simulator, but an explicit compiler analytical framework that formalizes interconnect transit, bank interleaving, and layout reorganization costs directly inside the compiler optimization decision pass.

2. Five Core Research & Technical Innovations

Innovation 1: Data-Movement-Aware Target Selection Matrix

Traditional compilers optimize strictly for operation latency and register allocation. NMC-Opt Studio elevates interconnect data traffic (MB transferred) and operational intensity (FLOP/Byte) to first-class compiler objective functions. The compiler autonomously computes the crossover scale (N_crossover) where memory bandwidth savings surpass PCIe/CXL bus transfer overheads.

Innovation 2: Unified Heterogeneous SSA Intermediate Representation (IR)

The framework introduces a specialized 3-address SSA IR that captures memory semantics across both host memory and distributed PIM banks. Standard IR instructions (load, store, fadd, gep) are augmented with bank-affinity annotations and explicit barrier synchronization instructions (sync_barrier) to expose physical bank interleaving to optimization passes.

Innovation 3: First-Principles Analytical Cost Engine with Interconnect Tax

Unlike slow cycle-accurate hardware emulators (gem5-PIM) that require hours for single benchmark runs, NMC-Opt Studio uses a closed-form first-principles analytical cost model. It factors in host DMA dispatch overheads (t_h2d, t_d2h), physical link bandwidth constraints (PCIe Gen5 vs CXL), and barrier latency, delivering instantaneous, cycle-representative compiler verdicts.

Innovation 4: In-Memory Layout Reorganization & Bank Striping Model

PIM speedup depends critically on avoiding bank channel contention. NMC-Opt Studio explicitly models the compile-time cost of converting contiguous host memory buffers into 16-bank interleaved layouts (T_layout), ensuring the compiler does not emit invalid offload decisions that assume unrealistic zero-cost data rearrangements.

Innovation 5: Context-Aware Compiler Copilot & Educational Workbench

The platform integrates an interactive compiler decision walkthrough with a retrieval-augmented Compiler Copilot. The assistant grounds real-time interactive parameter adjustments (e.g., sweeping vector scale or interconnect width) in compiler design principles and peer-reviewed PIM literature.

3. State-of-the-Art Comparative Matrix

The following matrix positions NMC-Opt Studio against prevailing software and simulation approaches:

Capability / Dimension	GCC / Clang	UPMEM SDK	gem5-PIM	NMC-Opt Studio
Target Selection	Host CPU only	Manual developer pragma	None (Trace driven)	**Autonomous Analytical Matrix**
Data Movement Model	Zero link cost	Manual host-device DMA	Cycle-level link sim	**Explicit First-Principles Tax**
Bank Interleaving	Ignored	Manual bank scatter	Hardware modeled	**Compile-time Layout Model**
Compilation Latency	Milliseconds	Seconds	Hours (Slow sim)	**Real-Time (<50 ms)**
Intermediate Rep.	Standard SSA	Clang LLVM IR	None (Sim traces)	**Bank-Aware 3-Address SSA**

4. Summary of Academic & Curricular Impact

NMC-Opt Studio establishes that future compiler architectures must evolve beyond pure instruction scheduling to incorporate interconnect and memory topology awareness. By providing an open, fully inspectable, and reproducible software workbench, the project enables both compiler researchers and students to explore heterogeneous memory-centric computing with mathematical rigor.